Element C: Presentation and Justification of Solution Design Requirements

Our school produces a lot of paper waste. A large majority of this waste comes from sheets of paper that have only been used or printed on one side. The other side is blank and completely usable, but the paper is getting thrown out anyway. This leads to more cost for the school since they have to pay more money for paper. It also led to more work for our housekeepers, who had to haul all of the extra paper waste out of the building and into dumpsters. The other paper waste comes from the composition notebooks that students use for classes. The cost of these notebooks comes directly out of each teacher's budget. However, in most cases, less than the first third of the notebook is used, and then it is not used again and just gets thrown out, even though the other two-thirds of the notebook has never been used.

The design requirements, prioritized by importance to our problem, are as follows:

1. The process to make the solutions is safe. This is a high priority. A safe process reflects that we are not using any dangerous objects, and that the final product is safe to touch. This is of high priority because if the process were dangerous or the product came out dangerous, there would not be anyone who would want our solution.
2. Each sheet has at least one side that is blank. This is a high priority. Each sheet having at least one side blank is a minimum for our design, making the solution actually usable.
3. Solutions cost less than $2.00 to produce. This is a high priority. This reflects the market price of the composition notebooks that our school bought. This is of high

priority because it makes our solution viable to produce.

4. Solutions are connected in some way. This is a high priority. This reflects how our team does not want a stack of just loose paper.
5. The solutions must have a size of 8.5”x11”, to 3x5, which is 3”x5”. This is a medium priority. This respects the convenience for our end users, as they would not use our product if it were too big or too small. This is of medium priority because it reflects a size for our solution that could fit into backpacks at the biggest and into most pockets at the smallest.
6. Reuse 90% of scrap paper from notebooks and misprints. This is a low priority. This requirement reflects a realistic goal for our project to not waste additional paper. This is low priority because we wanted a realistic percentage of paper that our team can reuse.
7. Solutions have at least 50 pages of paper. This is a low priority. We decided to go with 50 pages because we thought that that would be enough pages that would satisfy our stakeholders’ needs. This is low priority because the amount of pages can vary, and it would not affect our stakeholders that much.

These design requirements will lead to a solution that likely makes paper that has either one or two blank sides and is bound in some way into a notebook. Our stakeholders would be content with one-sided sheets of paper, but these design requirements will push us to create a better solution.

Mr. K, an engineering teacher at our school and one of our designated stakeholders, agreed with our high priority for safety, stating “Nobody wants to grab their notebook and get stabbed by a loose wire.” (Mr. K, personal communication, April 3, 2026). He also recommended to us the one side blank design requirement to high priority, stating that “This will work, but you also need to ensure there is nothing distracting or random notes on the other side of the sheet.” (Mr. K, personal communication, April 3, 2026). He also said that solutions being connected should also be high priority, stating that “If the notebook is not connected together, we are essentially just reusing random scrap paper.” (Mr. K, personal communication, April 3, 2026). He also helped clarify for us that our amount of pages need to be uniform, stating “They all need to have the same amount of sheets per notebook.” (Mr. K, personal communication, April 3, 2026).

One of our experts, Mr. H, an old engineering teacher that used to work at our school, also gave us feedback on our design requirements. Mr. H said “... did you consider as a high-priority design requirement the actual process of collecting the paper (who's going to do it) and the actual process of then redistributing the notebooks? Not to mention who's actually going to make the notebooks? Who's going to order the materials?” (Mr. H, personal communication, April 20, 2026)

References

Mr. K, personal communication, April 3, 2026

Mr. H, personal communication, April 20, 2026